

■ Liquid Morph UI — Technical Specification & Guide

Enterprise-Grade Glassmorphism & Behavior-Driven Liquid Glass UI Kit for Flutter (v1.0.3)

1. Executive Overview

Liquid Morph UI is a production-ready, high-performance Flutter package that delivers two distinct glass design paradigms: **Classic Glassmorphism** and **Dynamic Liquid Glass**. Built with Clean Architecture, rigid OOP design patterns, and hardware-accelerated 120 FPS GPU Canvas rendering, the package equips developers with modular AppBars, Bottom Navigation Bars, Switches, Buttons, Cards, and Text Fields.

Feature	Glassmorphism UI	Liquid Glass UI
Visual Depth	Static frosted depth with fixed 15px blur	Dynamic light refraction spot & cyan liquid glow
Motion Physics	Linear position interpolation	Viscous shape deformation & spring physics
GPU Acceleration	BackdropFilter + RepaintBoundary	Isolated CustomPainter + RepaintBoundary
Component Customization	Independent GlassAppBar & GlassCard	LiquidGlassAppBar, LiquidGlassSwitch & Buttons

2. Senior Developer Architecture & Design Patterns

The codebase enforces strict separation of concerns through classic Gang of Four (GoF) design patterns:

- **Clean Architecture (BLoC / Controller):** Business logic, touch gestures, and physics state are isolated inside GlassNavigationController.
- **Immutable State (GlassNavigationState):** Value objects with copyWith(), value equality (==), and hashCode prevent state mutation leaks.
- **Command Pattern (GlassNavigationCommand):** User actions are encapsulated as executable command objects (SelectTabCommand, DragUpdateCommand, DragEndCommand).
- **Scoped Dependency Injection (GlassScope):** Injects state controllers reactively down the BuildContext hierarchy using InheritedNotifier.
- **Strategy & Factory Patterns:** Standardized theme decorations created via GlassThemeFactory.

3. Glass UI Component Suite Specification

GlassAppBar & GlassAppBar.liquid

Standalone PreferredSizeWidget AppBars supporting custom title, titleWidget, leftButtons, rightButtons, background color, and blur.

GlassBottomNavigationBar & LiquidGlassBottomNavBar

Decoupled standalone bottom navigation bars accepting selectedIndex and onTap callbacks directly from any Scaffold.

GlassButton & LiquidGlassButton

Interactive buttons featuring frosted glass styling, press scaling animations, and active cyan glow shadows.

GlassSwitch & LiquidGlassSwitch

Animated toggle switches featuring frosted glass tracks and glowing thumb sliders.

GlassTextField & LiquidGlassTextField

Input fields with translucent frosted borders and focus glow shadows.

Widget Extensions (.asGlass() & .asLiquidGlass())

Convenient 1-line extension methods allowing ANY Widget to be wrapped in a frosted or liquid glass surface.

4. Quick-Start Code Examples

A. Standalone GlassAppBar & LiquidGlassBottomNavBar in Scaffold:

```
import 'package:flutter/material.dart'; import 'package:liquid_morph_ui/glass_ui_kit.dart';
Scaffold( backgroundColor: Colors.transparent, extendBody: true, extendBodyBehindAppBar: true,
appBar: GlassAppBar.liquid( title: 'Dashboard', rightButtons: [IconButton(icon:
Icon(Icons.settings), onPressed: () {})], ), body: pages[selectedIndex], bottomNavigationBar:
LiquidGlassBottomNavBar( selectedIndex: selectedIndex, onTap: (index) => setState(() =>
selectedIndex = index), items: myItems, ), );
```

B. Widget Extension Methods:

```
// Turn ANY widget into Frosted Glass in 1 line Text('Frosted Card').asGlass(blur: 15.0); // Turn
ANY widget into Liquid Glass in 1 line Text('Liquid Card').asLiquidGlass(glowColor:
Color(0x3038BDF8));
```

5. Package Publishing & Maintenance

To publish updates for **liquid_morph_ui** to **pub.dev**:

```
# 1. Update version in pubspec.yaml (e.g. version: 1.0.3) # 2. Add entry to CHANGELOG.md # 3.
Clean build cache & verify dry-run fvm flutter clean fvm flutter pub publish --dry-run # 4.
```

```
Upload live to pub.dev fvm flutter pub publish
```